

```
In [1]: import matplotlib.pyplot as plt
import numpy as np
import h5py
import os
from tqdm import tqdm
import glob
```

This is the note on how to load simulated Hinode data from the SPIn4D Data Release 2

First, get the path directory, for both simulation and stokes, and the index number

```
In [2]: path_stokes='/home/yangkai/koa_scratch/ToAndrew'
path_sim='/home/yangkai/koa_scratch/spin4d_bak/'
index='096480'
caseid='ssd100'
subdir='SPIN4D_SSD_100G'
```

```
In [3]: simulation_file=f'{path_sim}/{subdir}/spin4d-{caseid}-{index}-Hinode.h5'
stokes_file=f'{path_stokes}/{subdir}/spin4d-{caseid}-{index}-6302-stokes-Hir
```

Magnetic field unit is 3.5449077018110318 that is $\sqrt{4\pi}$

Temperature unit is K.

The variable "tau" is actually the geometric height in the unit of km.

Velocity unit is cm/s

```
In [4]: labels=['Bx', 'By', 'Bz', 'T', 'logP', 'tau', 'vz']
units=[np.sqrt(4*np.pi), np.sqrt(4*np.pi), np.sqrt(4*np.pi), 1, 1, 1, 1]
data={}
with h5py.File(simulation_file, 'r') as f:
    for l in tqdm(labels):
        data[l]=np.array(f[l])
    print(f.keys())
```

```
100%|██████████| 7/7 [00:00<00:00, 147.17it/s]
<KeysViewHDF5 ['Bx', 'By', 'Bz', 'T', 'logP', 'tau', 'vz']>
```

Attention: the Stokes axis order is incorrect.

The 1st axis is Y and the 2nd axis is X.

Swap the first two axes with `.transpose(1, 0, 2, 3)`.

```
In [5]: with h5py.File(stokes_file, 'r') as f:
        stokes=np.array(f['stokes']).transpose(1, 0, 2, 3)
        base_wl = f.attrs['lambda_zeropoint']
        wavelength=np.array(f['lambda'])+base_wl
        print(f.keys())
```

```
<KeysViewHDF5 ['lambda', 'stokes']>
```

Now the 1st is X, 2nd is Y.

3rd in simulation is Z.

3rd in Stokes is polarization state.

4th in Stokes is wavelength

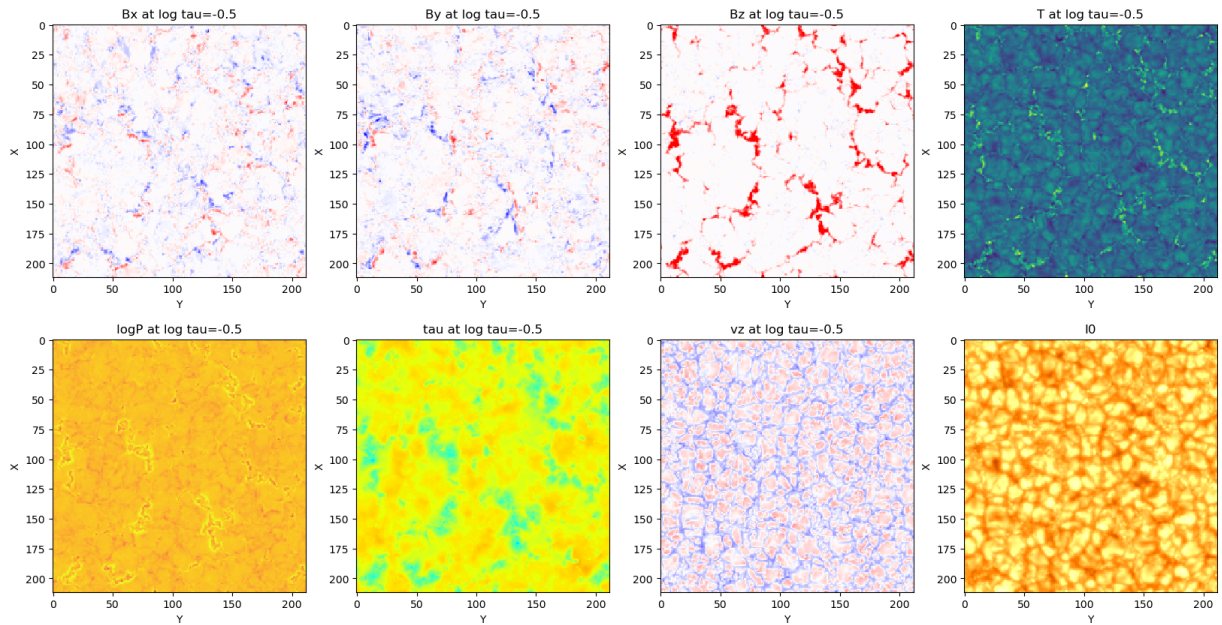
```
In [6]: ltaiu_final = np.linspace(1.0, -4, 11)
```

base_wl is the base value of the wavelength, thus the absolute wavelength should be `f['lambda']+base_wl`

The atmospheric parameters are sampled at multiple optical depth, from 1 to -4 with 11 sample points.

```
In [7]: fig,axes=plt.subplots(2,4,figsize=(20,10))
        axes=axes.flatten()
        ih=3
        cmaps=['bwr','bwr','bwr','viridis','plasma','jet','bwr','afmhot']
        vmaxs=[2e2,2e2,5e2,6500,5,1600,1e6,1.2]
        vmis=[-2e2,-2e2,-5e2,5000,4,300,-1e6,0.5]

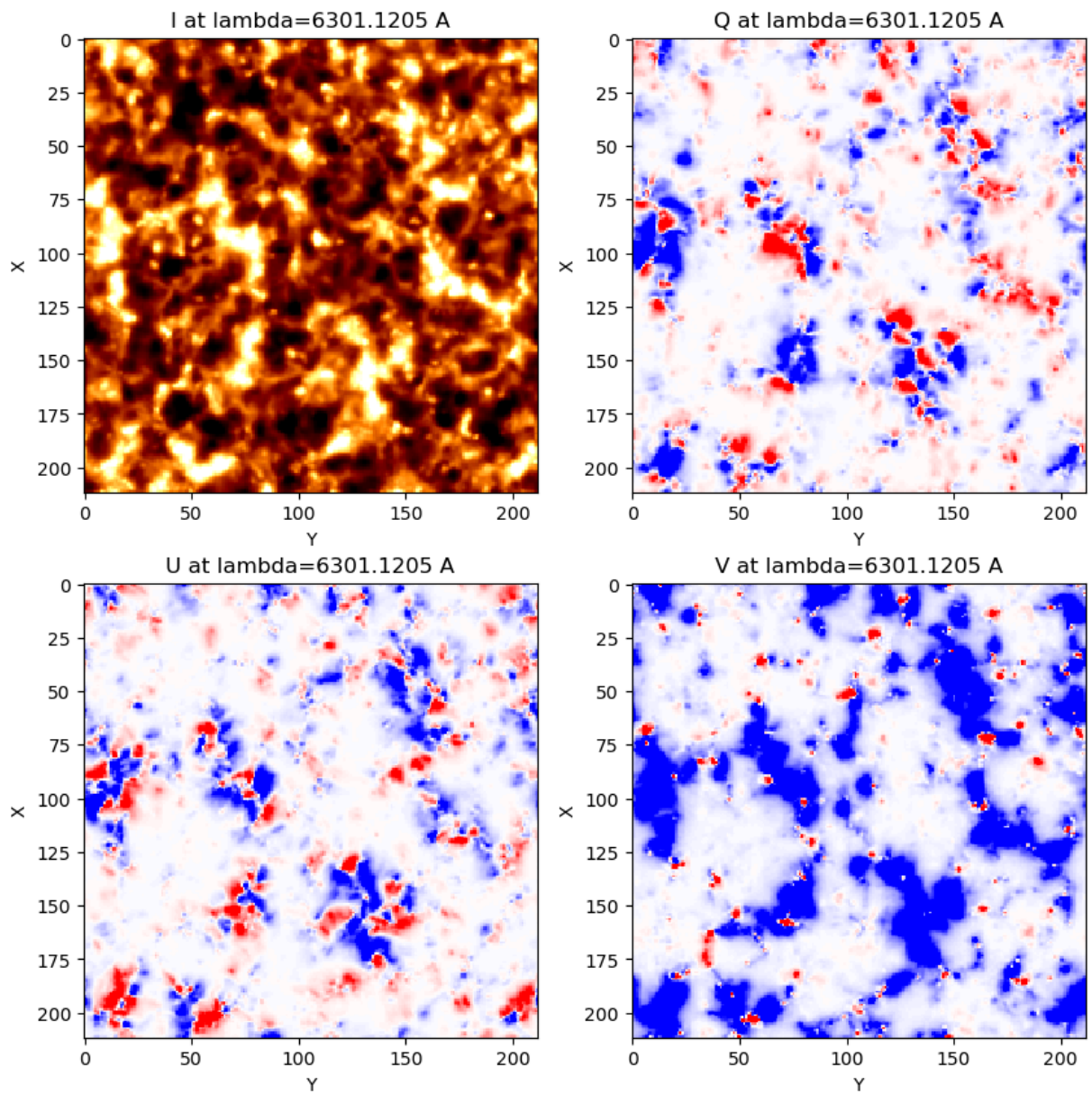
        for idx in range(8):
            axes[idx].set_xlabel('Y')
            axes[idx].set_ylabel('X')
            if idx < 7:
                axes[idx].imshow(data[labels[idx]][...,ih], cmap=cmaps[idx], vmax=vmaxs[idx], vmin=vmis[idx])
                axes[idx].set_title(f'{labels[idx]} at log tau={ltaiu_final[ih]}')
            else:
                axes[idx].imshow(stokes[...,0,0], cmap=cmaps[idx], vmax=vmaxs[idx], vmin=vmis[idx])
                axes[idx].set_title('I0')
```



```
In [8]: fig, axes = plt.subplots(2, 2, figsize=(10, 10))
        axes = axes.flatten()
        iw = 30
        cmaps = ['afmhot', 'bwr', 'bwr', 'bwr']
        vmaxs = [0.5, 1e-3, 1e-3, 1e-2]
        vmis = [0.2, -1e-3, -1e-3, -1e-2]
        stokesid = ['I', 'Q', 'U', 'V']

        for idx in range(4):
            axes[idx].set_xlabel('Y')
            axes[idx].set_ylabel('X')

            axes[idx].imshow(stokes[... , idx, iw], cmap=cmaps[idx], vmax=vmaxs[idx], vmis=vmis[idx])
            axes[idx].set_title(f'{stokesid[idx]} at lambda={wavelength[iw]*1e-3:.4f}
```



In []: